AUTONOMOUS ROBOTIC DATA LOGISTICS SYSTEM

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Title: AUTONOMOUS ROBOTIC DATA LOGISTICS SYSTEM

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Cross-Reference to Related Applications

This application claims the benefit of and priority to U.S. Patent Application No. 19/559,705, filed on or about March 6, 2026, the contents of which are hereby incorporated by reference in their entirety.

Abstract

A system for the automated physical transport of data, comprising an autonomous robotic unit equipped with a manipulator and an interface configured to handle a portable data cartridge having a high-bandwidth optical interface. The system is designed to solve the "last mile" or "last meter" problem in a physical data logistics network. The robotic unit, which may be an aerial drone, a ground vehicle, or a stationary robotic arm, can autonomously retrieve a data cartridge from a data port, transport it to a different location, and insert it into a destination port. This enables the creation of fully automated, dynamically reconfigurable data fabrics within data centers or the rapid deployment of massive datasets between remote locations without human intervention.

Background of the Invention

While the development of data cartridges with high-bandwidth optical interfaces makes the physical transport of data viable, the process of moving these cartridges still relies on manual intervention. In a data center environment, this means technicians must physically move cartridges between servers. For remote data collection, it requires personnel to travel to the sensor location to retrieve the data. This manual handling introduces delays, labor costs, and potential for human error. A system is needed to fully automate the physical logistics of data transport at a local level.

Summary of the Invention

The present invention is a system that combines an autonomous robotic unit with a specialized portable data cartridge having a high-bandwidth optical interface to create a fully automated data logistics platform.

The system comprises an autonomous vehicle, such as an unmanned aerial vehicle (drone), a self-driving ground vehicle, or a robotic arm, which is communicatively coupled to a central command system. The key feature is that the robotic unit is equipped with a specialized manipulator and an optical data interface designed to physically dock with and handle a portable data cartridge.

In one embodiment, termed the "Optically-Attached Drone" (OAD), a drone is dispatched to a remote sensor or facility. It lands, uses its manipulator to retrieve a data cartridge full of new data from a port, and transports it back to a central processing center, where it deposits the cartridge for high-speed offload. In another embodiment, robotic arms within a data center continuously reconfigure the data storage architecture by physically moving cartridges between different servers or storage arrays, creating a "physical data fabric" that can be optimized in real-time.

Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Federal Government and its military and defense agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental,

defense, and non-commercial purposes. This license explicitly excludes any and all state and local governments.

Brief Description of the Drawings

FIG. 1 is a conceptual diagram of one embodiment, the Optically-Attached Drone (OAD), retrieving a data cartridge from a remote station.

FIG. 2 is a conceptual diagram of another embodiment, showing robotic arms reconfiguring a data center by moving data cartridges between servers.

Detailed Description of the Invention

Referring to **FIG. 1**, the Optically-Attached Drone (OAD) embodiment of the system is shown. An autonomous drone (101) is equipped with a specialized manipulator arm (102) and an optical data interface. The drone is dispatched to a remote data collection outpost (103), which has a data port (104). The drone lands and aligns itself with the port, uses its manipulator (102) to grasp and retrieve a portable data cartridge (105) from the port, and then transports the cartridge to a destination.

Referring to **FIG. 2**, the data center embodiment is shown. A series of server racks (201) are equipped with optical data ports (202) that can accept portable data cartridges. Instead of being connected by a fixed network of cables, the data storage is mobile. Autonomous robotic arms (203) are mounted on a rail system (204). Under the direction of a data center management system, the arms can retrieve a data cartridge (105) from one server's port and physically transport and insert it into the port of another server, thereby reconfiguring the data storage topology of the data center in real-time.

Claims

What is claimed is:

1. A system for automated data logistics, comprising:

    - a portable data cartridge having a massively parallel optical interface; and

- an autonomous robotic unit comprising a manipulator configured to physically grasp, transport, and release the data cartridge.

2. The system of claim 1, wherein the robotic unit is an unmanned aerial vehicle (drone).

3. The system of claim 1, wherein the robotic unit is a robotic arm installed within a data center.

4. The system of claim 1, wherein the robotic unit is further configured to dock with a data port and facilitate the transfer of data to or from the data cartridge via the optical interface.

5. A method for physically transporting data, comprising the steps of:

   - dispatching an autonomous robotic unit to a first location having a data port containing a portable optical data cartridge;
   - causing the robotic unit to use a manipulator to retrieve the data cartridge from the port;
   - transporting, by the robotic unit, the data cartridge to a second location; and
   - causing the robotic unit to deposit the data cartridge at the second location.

Drawings

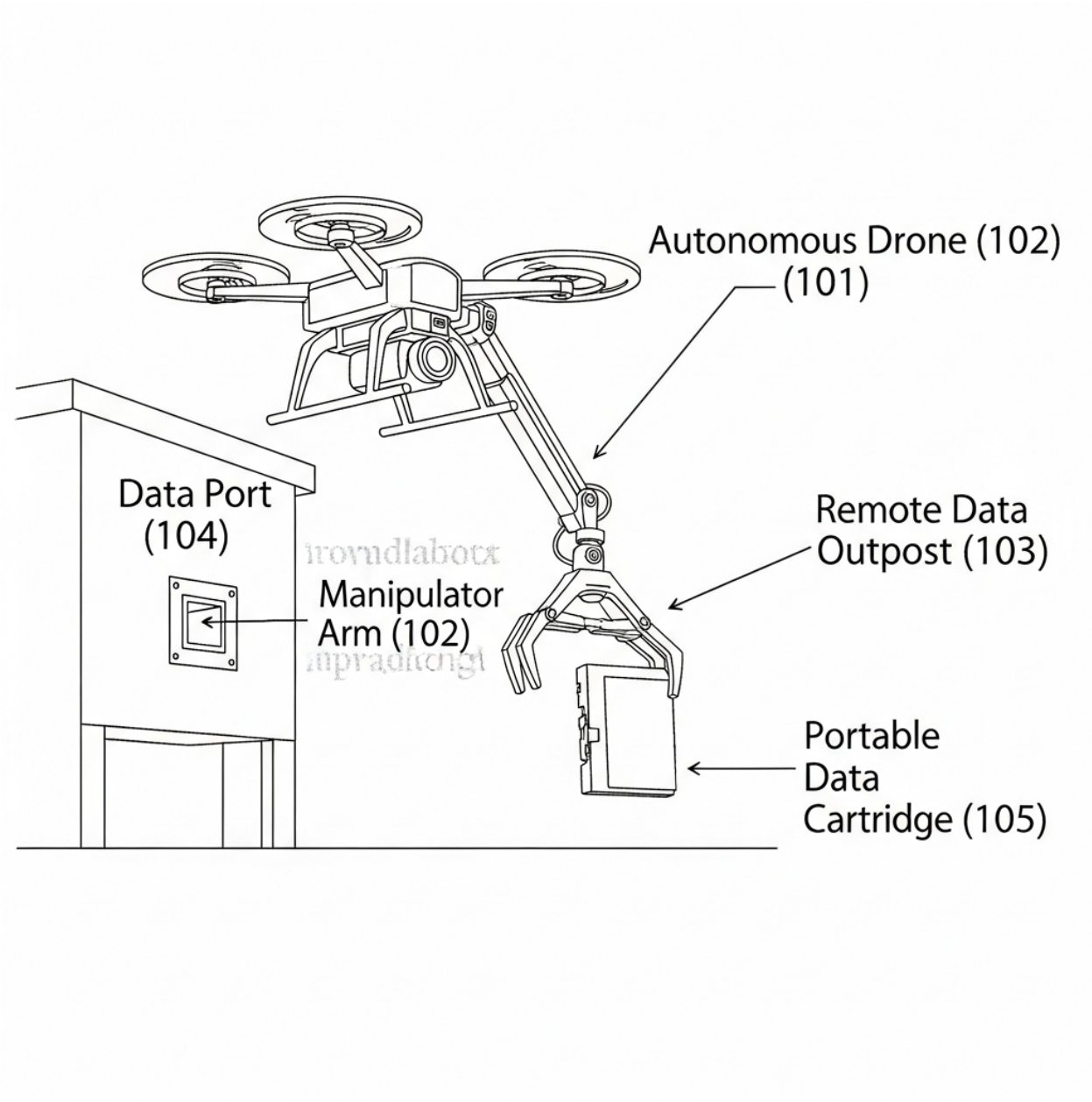


Figure 1: A conceptual diagram of one embodiment, the Optically-Attached Drone (OAD), retrieving a data cartridge from a remote station.

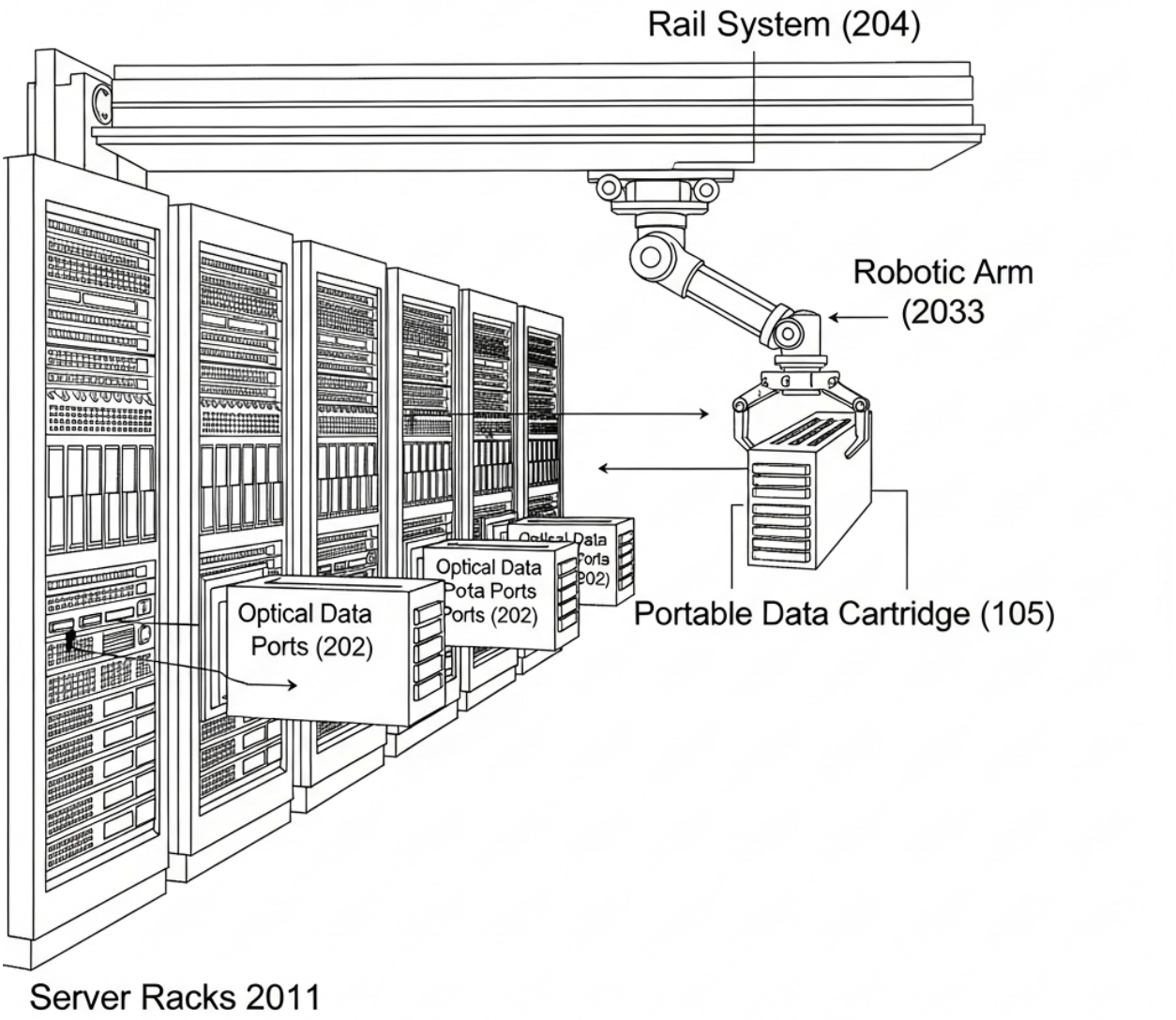


Figure 2: A conceptual diagram of another embodiment, showing robotic arms reconfiguring a data center by moving data cartridges between servers.